



Working with Data Analysis and Relational Databases **in Python**



Every day, we generate a huge volume of varied data, which in its raw state, does nothing but occupy space. However, if the data is organised and manipulated, meaningful and valuable conclusions can be drawn from it. Therefore, the hunt is always on for tools that assist in the process of handling data. Python Pandas is one such tool.

Data science is the process of deriving insights from a huge and diverse set of data by organising, processing and analysing it. This practice is prevalent in many different domains like e-commerce, healthcare and energy. The programming requirement of data science demands a very versatile yet flexible language, the code for which is simple to write but which can handle highly complex mathematical processing. Python is most suited for such requirements as it is robust enough to handle complex scenarios with minimal code and less confusion. It has a very large collection of libraries that serve as special-purpose analysis tools.

Pandas is an open source Python library used for high-performance data manipulation and data analysis, using its powerful data structures. It is suitable for many different types of data, including table data with heterogeneous

columns, such as SQL tables; or Excel data, ordered and unordered time series data, matrix data with row and column labels, or other forms of observation/statistical data sets. Python with Pandas is used in various academic and commercial domains, including economics, statistics, advertising, Web analytics, and more.

This article briefly covers data processing with Python and connectivity to relational databases using Python. It includes queries that allow data manipulations within the Pythonic framework provided by the Python SQL toolkit, SQLAlchemy, which is said to be the best Python database abstraction tool as it provides an accessible way to query, build and write to the SQLite database.

Python: Processing CSV data

Reading data from CSV (comma separated values) is a